10. **Photosynthesis occurs in –**

(a) Night (b) Day and night

(c) Day or night (d) Only day

**43rd B.P.S.C. (Pre) 1999**

**Ans. (c)**

Through the process of photosynthesis, green plants have a capacity of manufacturing their food from simple substances as $CO_2$ and $H_2O$ in presence of light. Normally, plants utilize sunlight (day) but marine algae also use moonlight. Photosynthesis even occurs in electric light.

11. **The process by which plants produce food is called :**

(a) Carbohydrolysis (b) Metabolic synthesis

(c) Photosensitization (d) Photosynthesis

**44th B.P.S.C. (Pre) 2000**

**Ans. (d)**

See the explanation of the above question.

12. **Which metal ion is found in chlorophyll?**

(a) Iron (b) Magnesium

(c) Zinc (d) Cobalt

**46th B.P.S.C. (Pre) 2004**

**Ans. (b)**

Magnesium ion is found in chlorophyll, chlorophyll is the green photosynthetic pigment present in plants. It initiates the process of converting light energy into chemical energy through the process of photosynthesis.

13. **Photosynthesis occurs in-**

(a) Nucleus (b) Mitochondria

(c) Chloroplast (d) Peroxisome

**53rd to 55th B.P.S.C. (Pre) 2011**

**Ans. (c)**

Photosynthesis takes place primarily in leaves and partially in stems. It takes place within especialized cell structures called chloroplasts. A leaf has a petiole or the stalk and a lamina, the flat portion of the leaf. As its area is broad, the lamina helps in the absorption of sunlight and carbon dioxide during photosynthesis. Photosynthesis takes place in the chloroplasts that have chlorophyll present in them. It is the chlorophyll that absorbs light energy from the sun. There are tiny pores called stomata that function as roadways for carbon dioxide to enter and oxygen to leave the plant.

14. **Solar energy is converted into ATP in :**

(a) Mitochondria (b) Chloroplast

(c) Ribosome (d) Peroxisome

(e) None of the above/More than one of the above

**65th B.P.S.C. (Pre) 2019**

**Ans. (b)**

In the process of photosynthesis solar energy is converted into ATP in chloroplast. In light reactions (energy-transduction reactions) of photosynthesis solar energy is converted into chemical energy in the form of two energy-transporting molecules, ATP and NADPH.

15. **Which among the following is a character of chloroplast which makes them qualified to self-replication?**

(a) Presence of both DNA and RNA

(b) Presence of DNA only

(c) Absence of RNA

(d) More than one of the above

(e) None of the above

**68th B.P.S.C. (Pre) 2022**

**Ans. (a)**

Presence of DNA and RNA both are a character of chloroplast makes them qualified to self-replication. Chloroplasts, the organelles responsible for photosynthesis are in many respects similar to mitochandria. Plant chloroplasts are like mitochandria are bounded by double membrane called the chloroplast envelope. In addition to this membrane chloroplasts have a third membrane called the Thylakoid membrane. The major difference between chloroplasts and mitochandria in terms of both structure and function is the Thylakoid membrane.

16. **Which pigment is essential for nitrogen fixation by leguminous plants?**

(a) Anthocyanin (b) Leghaemoglobin

(c) Phycocyanin (d) More than one of the above

(e) None of the above

**B.P.S.C. School Teacher (Class 6-8) 09-12-2023**

**Ans. (b)**

The pigment essential for nitrogen fixation by leguminous plants is called "leghemoglobin". It is a protein found in the root nodules of leguminous plants like soyabeans, peas etc. Leghemoglobin helps in creating an oxygen-free environment within the root nodules, which is crucial for the functioning of nitrogenase, the enzyme responsible for nitrogen fixation.

# II. Plant Nutrition

1. **Water is conducted in vascular plants by-**

(a) Phloem tissue (b) Parenchyma tissue

(c) Meristems (d) Xylem tissue

**53rd to 55th B.P.S.C. (Pre) 2011**

**Ans. (d)**

The xylem transports water and soluble mineral nutrients from the roots throughout the plant. It is also used to replace water lost during transpiration and photosynthesis.

2. **Water in plants is transported by :**

(a) Xylem (b) Epidermis

(c) Phloem (d) Cambium

(e) None of the above/More than one of the above

**66th B.P.S.C. (Pre) (Re. Exam) 2020**

**Ans. (a)**

There are two types of transporting tissues found in vascular plants. (i) Phloem and (ii) Xylem. Xylem is the specialised tissue of vascular plants that transports water and minerals from plants. Soil interface to stems and leaves and provides mechanical support and storage.

3. **Water passes from the soil into the root by the physical process is called -**

(a) Diffusion (b) Transpiration

(c) Absorption (d) Osmosis

(e) None of the above / more than one of the above

**67th B.P.S.C. (Pre) 2021**

**Ans. (d)**

Water reaches the roots from the soil through the process of osmosis.

4. **Water enters from soil into root hairs of plants due to**

(a) atmospheric pressure (b) capillarity

(c) osmotic pressure (d) More than one of the above

(e) None of the above

**B.P.S.C. Headmaster 07-12-2023**

**Ans. (c)**

Water enters the root hairs by the process of osmosis. Osmosis is the process by which the molecules of a solvent, such as water, move from a region of its higher concentration to a region of its lower concentration through a semipermeable membrane. This decreases the water potential inside the cell, so water molecules move down gradient into the cell.

5. **Which of the following is present in the intestine of termites to digest cellulose?**

(a) Protozoa (b) Yeast

(c) Bacteria (d) Algae

(e) None of the above / More than one of the above

**B.P.S.C. Auditor 2021**

**Ans. (a)**

Termites have cellulolytic enzyme in their gut to digest the cellulose. The protozoa break the cellulose into the simple sugar thereby termites are able to metabolize the cellulose. Termites eat wood, which is mostly cellulose which they can not digest without their symbiote gut protozoans.

6. **Water reaches great heights in trees because of suction pull caused by**

(a) evaporation (b) absorption

(c) transpiration (d) More than one of the above

(e) None of the above

**68th B.P.S.C. (Pre) 2022**

**Ans. (c)**

Water reaches great heights in trees because of suction pull caused by Transpiration.
Transpiration is the process in which water is lost as water vapor from the aerial parts of the plant through stomata.
Transpiration is essential evaporation of water from the leaves of the plant.
Transpiration is ultimately the main driver of water movement in Xylem.
Transpiration is a passive process, meaning that ATP is not required for water movement.

7. **Pneumatophores are present in**

(a) hydrophytes (b) xerophytes

(c) halophytes (d) More than one of the above

(e) None of the above

**B.P.S.C. PRT 2023 Shift II**

**Ans. (d)**

In some plants such as Rhizophora growing in swampy area, many roots come out of the ground and grow vertically upwards. Such roots called pneumatophores, help to get oxygen for respiration. Pneumatophores are present in hydrophytes, halophytes plants.

8. **In how many steps $CO_2$ is released in aerobic respiration?**

(a) One (b) Six

(c) Three (d) More than one of the above

(e) None of the above

**B.P.S.C. School Teacher (Class 6-8) 10-12-2023**

**Ans. (c)**

Aerobic respiration is the process that leads to a complete oxidation of organic substances in the presence of oxygen, and releases $CO_2$, water and a large amount of energy present in the substrate. This type of respiration is most common in higher organisms. In three steps $CO_2$ is released in aerobic respiration. Those steps are:

- During acetyl CO - A formation
- During formation of alpha-Ketoglutaric acid in krebs cycle
- During succinic acid formation in krebs cycle.